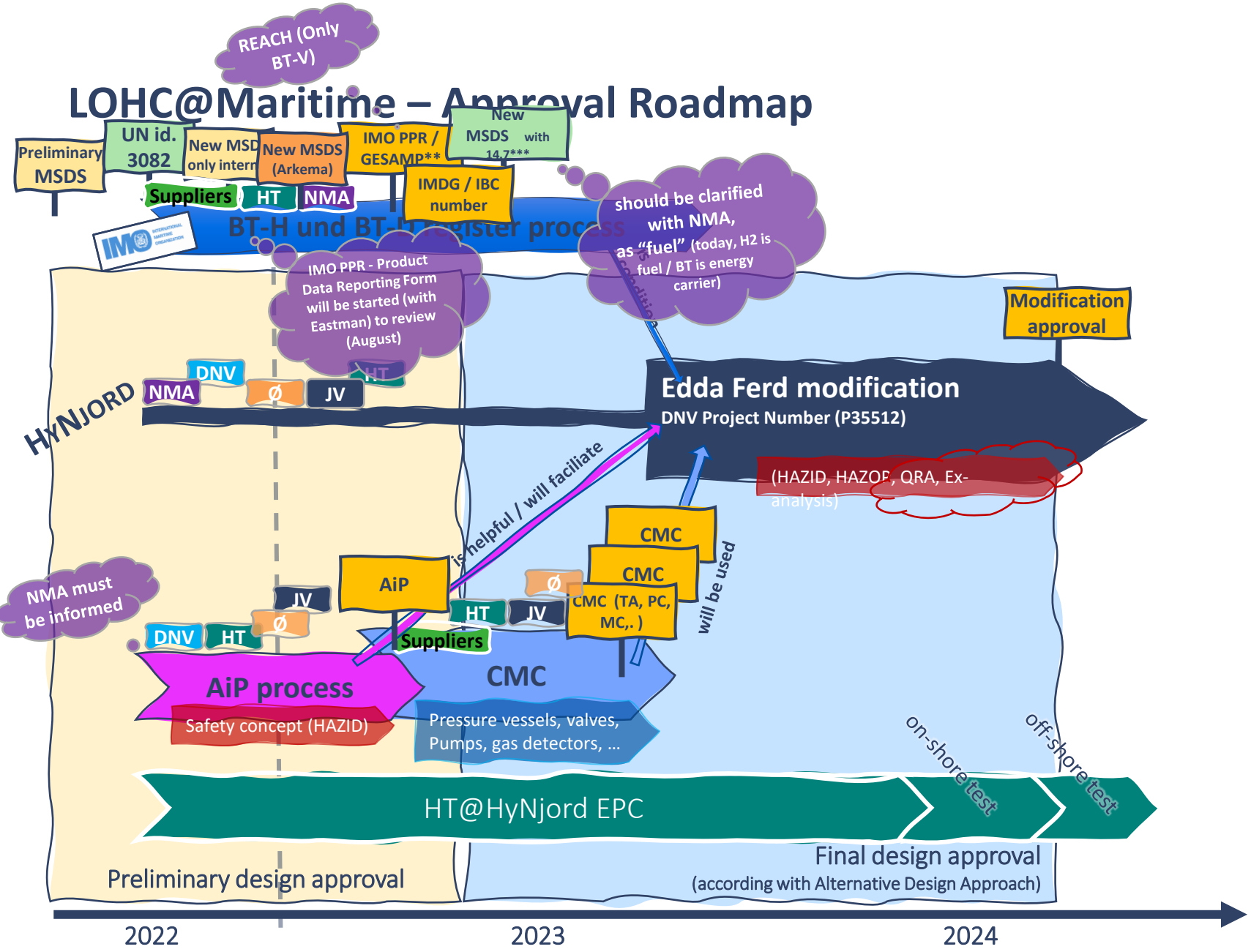
LOHC@Maritime – Approval Roadmap



Work in Progress

CONTACTS

- DNV for Approval in Principle (AiP)
  - Sergey Grunda
  - Dalibor Bukarica
- DNV for Edda Ferd Modification
  - Eirik Lindseth (Project Manager)
- Norwegian Maritime Authority (NMA)
  - Raymond Lean
  - Andrea Skarstein
  - Karolina Lundgren
  - Elisabeth Førland

- Approval in Principle (AIP)
- Certification of Materials and Components (CMC)
- Quantitative Risk Assessment (QRA)
- United Nations id or number (UN id)
- Material Safety Data Sheet (MSDS)
- Norwerigian Maritime Authority NMA
- Der Noske Veritas Class DNV
- Joint Venture JV
- Østensjø Rederi Ø
- Hydrogenious LOHC Technologies HT
- Properties and characteristics of products proposed for bulk marine transport (PPR Product Data Reporting Form)

** There is currently no GESAMP hazard profile (GHP) for the substance, you as the manufacturer will need to assign a provisional hazard profile (GHP).
*** Transport in bulk according to Annex II of MARPOL 73/78 and the IBC Code

on-board

LOHC@Maritime – Approval Roadmap

maritime regulations on dissolved gases in fuels?

- SOLAS convention - Chapter VI - Carriage of cargoes and oil fuels (https://www.imorules.com/SOLAS_REGVI.html)
- SOLAS convention - Chapter VII - Carriage of dangerous goods (https://www.imorules.com/SOLAS_REGVII.html)
- Annex II of MARPOL 73/78 Regulations for the Control of Pollution by Noxious Liquid Substances in Bulk convention (https://www.imorules.com/MARPOL_ANNII.html)
- *Both SOLAS – Ch.VII & MARPOL Annex II conventions require all chemical tankers should comply with IBC Code (International Bulk Chemical Code).

Work in Progress

Arkema will complete for us.

IMO PPR Product Data Reporting Form must be started for BT

MSDS and/or other the available information about the substance and its hazard levels or concentrations.

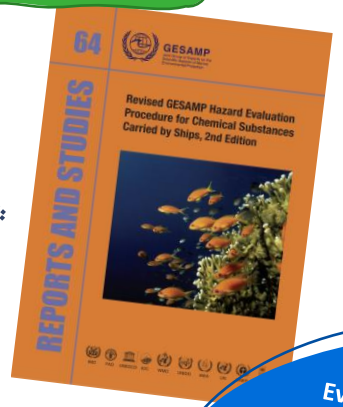
Section 5 - GESAMP Hazard Profile

Currently, no GESAMP hazard profile (GHP) for

Suppliers

will need to assign a provisional hazard profile (GHP).

Appendix 1 of MARPOL Annex II (tables 2,3,4 with columns A,B, C, D and E).



Every ship certified to carry substances of IMO MARPOL Annex II Category X, Y, or Z should have onboard a manual approved by Administration and be certificated to carry that product.

2022 → 2023

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Certifications of materials, components and systems

The objective of certification is to ensure that **materials, components and systems used in** [illegible] be classed by the Society [illegible] uirements.

Certification of materials, components and systems (CMC) defined in **DNV-CG-0550 Sec.3**.

Certification normally includes both **plan approval** and **survey** during production and/or of the final product
- Plan approval
- Type approval
- Survey or (MSA)

Plan approval

The **plan approval of materials, components and systems** shall be:
1. on a **case by case basis** or
2. follow the procedure for **type approval**

Letter of approval

A **letter of approval** or design verification report when compliance with the requirements for the design has been confirmed.

The results of the assessment will be stated in a letter of approval

Type approval

Type approval is a procedure for approval of standard designs and/or routinely manufactured, identical components to be used in DNV classed objects.
The type approval procedure may consist of the following elements:
— design assessment
— initial survey
— type testing
— issue of a type approval certificate
* The type approval procedure used by the Society is described in **DNV-CP-0338**.

For certain products and systems, as defined in applicable rules:
1. only type approval is required (For these products and systems **no survey is required**, i.e. **no product certificate is required**.)
2. type approval is a **mandatory procedure** for **plan approval**.

Certification requirements for manufacturers

- **Type approval certificate (TAC)**

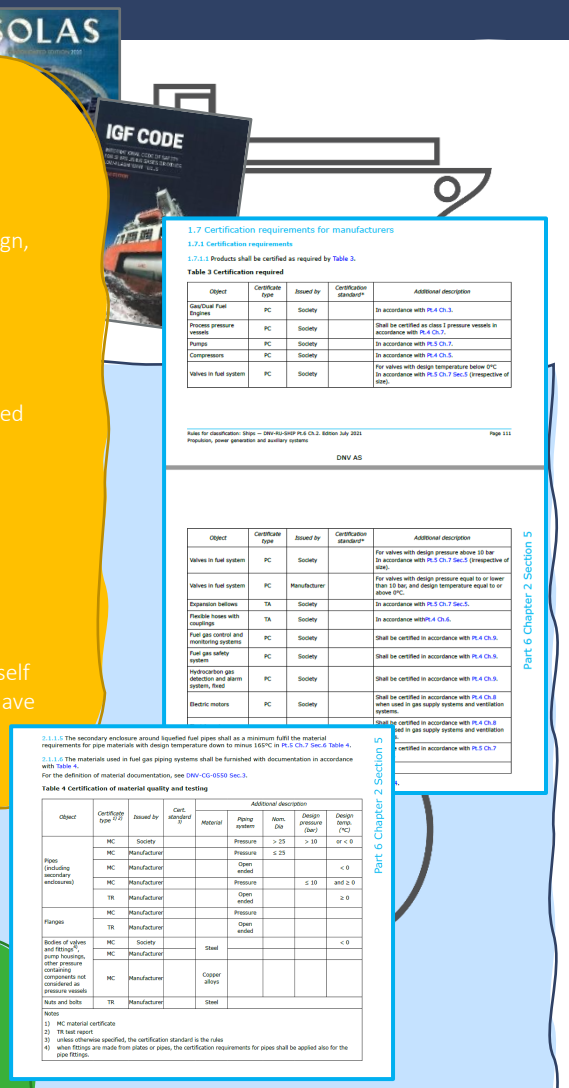
A certificate relating to and identifying a product design, based upon:
— design assessment
— production assessment, and/or
— survey of at least one product.

- **Product certificate (PC)**

A certificate relating to and identifying a product, based upon:
— design assessment
— survey of product and/or production.

- **Material certificate (MC)**

A certificate
— relating to and identifying a product
— referring to material related requirements of the referred standards
— based upon survey performed upon the product itself
— stating the results of tests and examinations that have been performed upon the product
- …



Type approval certificate states that the design of a product type is in conformity with specified requirements.
Product certificate issued that a manufactured product is in conformity with specified (design and production) requirements.

Final design

All materials, components and systems used in vessels to be classed by the Society.

Certifications of materials, components and systems

The objective of certification is to ensure that **materials, components and systems used in vessels** to be classed by the Society comply with the rule requirements.

Certification normally includes both **plan approval** and **survey** during production and/or of the final product

Material and Equipment certification

- Conformity Assessment Service
- Type Approval
- Product Design Assessment
- Individual Product Manufacturing
- Third-Party Certification Services

Compliance with rules for materials, systems and components will be confirmed by the types of documents defined in DNV-CG-0550 Sec.3.

Mechanical component documentation

The documentation shall be based on a relevant selection from the following documentation types:

— C010 Design criteria
— C020 Assembly or arrangement drawing
— C030 Detailed drawing
— C040 Design analysis
— E090 Table of Ex-installation
— M010 Material specification, metals
— M150 Non-destructive testing (NDT) plan
— S090 Specification of piping, valves, flanges and fittings.
as applicable for the component.

Class guideline — DNV-CG-0550. Edition September 2021

The **plan approval of materials, components and systems** shall be:

1. on a case by case basis or
2. follow the procedure for **type approval**

A **letter of approval** or design verification report when compliance with the requirements for the design has been confirmed.

The type approval procedure may consist of the following elements:

— design assessment
— initial survey
— type testing
— issue of a type approval certificate

The following shall be submitted for approval

— documentation of the design
— results of type testing normally witnessed by a surveyor.

For certain products and systems, as defined in applicable rules:

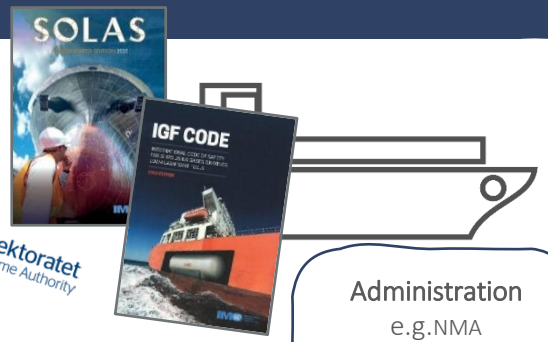
1. only type approval is required (For these products and systems **no survey is required**, i.e. **no product certificate is required**.)
2. type approval is a **mandatory procedure** for plan approval.

Final design

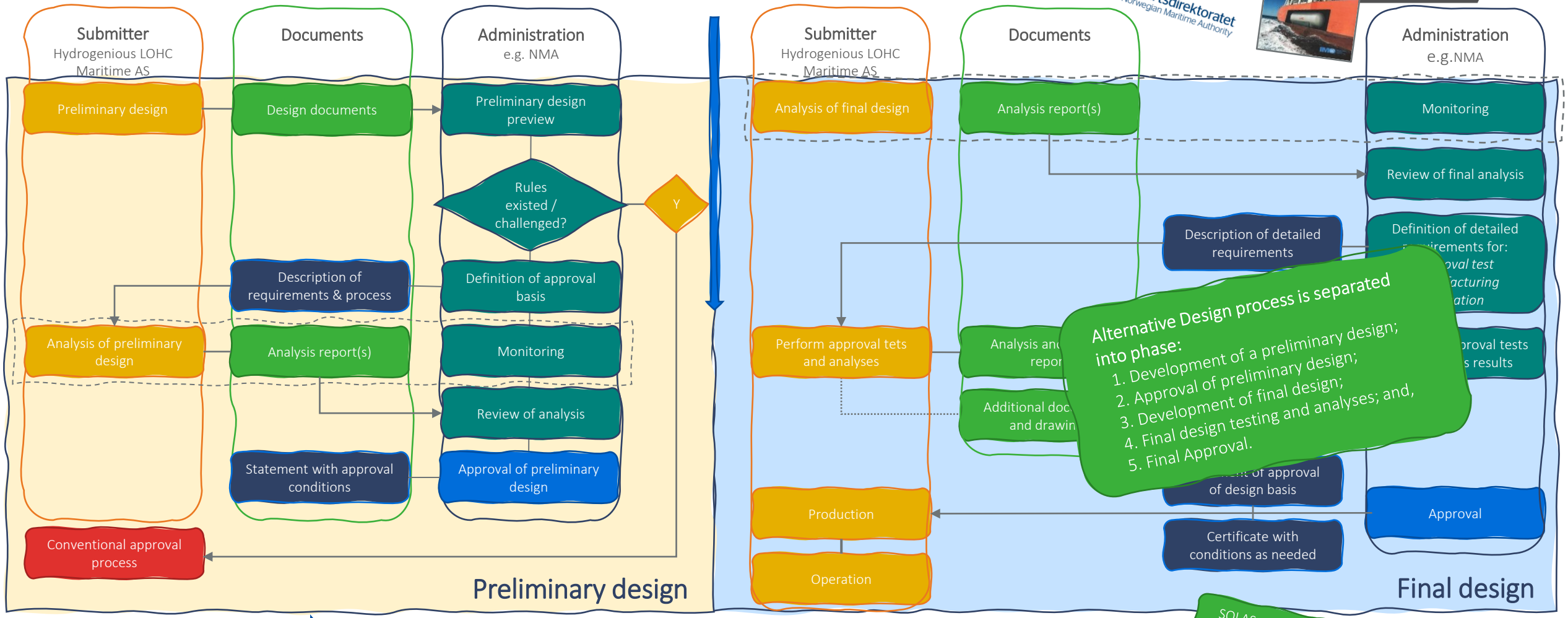
All materials, components and systems used in vessels to be classed by the Society.

Alternative Design Approach opens for solutions not covered by prescriptive rules

Process for non-regulated fuels or systems



Sjøfartsdirektoratet
Norwegian Maritime Authority



Alternative Design process is separated into phase:
1. Development of a preliminary design;
2. Approval of preliminary design;
3. Development of final design;
4. Final design testing and analyses; and,
5. Final Approval.

SOLAS regulation II-55
MSC.1/Circ.1455

The **Alternative Design approach** is the approval process required from the **flag authority** for some **novel designs**, e.g. Hydrogenious technology, which applies to projects that are already in a newbuilding pathway and is **done by the Flag in cooperation with the class society**. The goal is to obtain approval from the Flag Authority that is used by the ship and demonstrate compliance with the IMO regulations

IMO INTERNATIONAL MARITIME ORGANIZATION

on-board

Notes

- CMC = Certification of Materials and Components
- QRA = Quantitative Risk Assessment
- AIP = Approval in Principle
- GASA = General Approval in Ship Application

1.7 Certification requirements for manufacturers

1.7.1 Certification requirements

1.7.1.1 Products shall be certified as required by Table 3.

Table 3 Certification required

Object	*Certificate type*	*Issued by*	*Certification standard**	*Additional description*
Gas/Dual Fuel Engines	PC	Society		In accordance with Pt.4 Ch.3.
Process pressure vessels	PC	Society		Shall be certified as class I pressure vessels in accordance with Pt.4 Ch.7.
Pumps	PC	Society		In accordance with Pt.5 Ch.7.
Compressors	PC	Society		In accordance with Pt.4 Ch.5.
Valves in fuel system	PC	Society		For valves with design temperature below 0°C In accordance with Pt.5 Ch.7 Sec.5 (irrespective of size).

2.1.1.5 The secondary enclosure around liquefie[illegible]l pipes shall as a minimum fulfil the material requirements for pipe materials with design temp[illegible]ure down to minus 165°C in Pt.5 Ch.7 Sec.6 Table 4.

2.1.1.6 The materials used in fuel gas piping systems shall be furnished with documentation in accordance with Table 4.

For the definition of material documentation, see DNV-CG-0550 Sec.3.

Table 4 Certification of material quality and testing

Object	*Certificate type 1) 2)*	*Issued by*	*Cert. standard 3)*	*Additional description*				
				Material	*Piping system*	*Nom. Dia*	*Design pressure (bar)*	*Design temp. (°C)*
Pipes (including secondary enclosures)	MC	Society			Pressure	> 25	> 10	or < 0
	MC	Manufacturer			Pressure	≤ 25		
	MC	Manufacturer			Open ended			< 0
	MC	Manufacturer			Pressure		≤ 10	and ≥ 0
	TR	Manufacturer			Open ended			≥ 0
Flanges	MC	Manufacturer			Pressure			
	TR	Manufacturer			Open ended			
Bodies of valves and fittings[4], pump housings, other pressure containing components not considered as pressure vessels	MC	Society		Steel				< 0
	MC	Manufacturer						
	MC	Manufacturer		Copper alloys				
Nuts and bolts	TR	Manufacturer		Steel				

Notes
1) MC material certificate
2) TR test report
3) unless otherwise specified, the certification standard is the rules
4) when fittings are made from plates or pipes, the certification requirements for pipes shall be applied also for the pipe fittings.

Part 6 Chapter 2 Section 5

Object	*Certificate type*	*Issued by*	*Certification standard**	*Additional description*
Valves in fuel system	PC	Society		For valves with design pressure above 10 bar In accordance with Pt.5 Ch.7 Sec.5 (irrespective of size).
Valves in fuel system	PC	Manufacturer		For valves with design pressure equal to or lower than 10 bar, and design temperature equal to or above 0°C.
Expansion bellows	TA	Society		In accordance with Pt.5 Ch.7 Sec.5.
Flexible hoses with couplings	TA	Society		In accordance withPt.4 Ch.6.
Fuel gas control and monitoring systems	PC	Society		Shall be certified in accordance with Pt.4 Ch.9.
Fuel gas safety system	PC	Society		Shall be certified in accordance with Pt.4 Ch.9.
Hydrocarbon gas detection and alarm system, fixed	PC	Society		Shall be certified in accordance with Pt.4 Ch.9.
Electric motors	PC	Society		Shall be certified in accordance with Pt.4 Ch.8 when used in gas supply systems and ventilation systems.
Electric motor starters	PC	Society		Shall be certified in accordance with Pt.4 Ch.8 when used in gas supply systems and ventilation systems.
Ventilation fans for hazardous spaces	PC	Society		Shall be certified in accordance with Pt.5 Ch.7 Sec.12.

*Unless otherwise specified the certification standard is DNV rules.

1.7.1.2 For general certification requirements, see DNV-CG-0550 Sec.4.

Part 6 Chapter 2 Section 5

Notes

- **Plan Approval -** A systematic and independent assessment of drawings, design documents or records by the Society to verify compliance with the rules or statutory requirements where authorised by a flag administration.
  - The results of assessment will be stated in a **letter of approval**.
  - Where **subcontractors and suppliers** are involved, the **customer shall co-ordinate the submission of required documentation**, as well as co-ordinate any approval comments given by the Society.
- **DNV approval of manufacturers (AoM)** confirms that you have been found in compliance with international standards, regulations or our recognised DNV rules. This provides your company with proof of high quality and safety standards.
- <u>Certification of materials and components</u>
  - These include certification of materials, components and systems relevant to the safe operation of ships, offshore installations, …
  - Range of technical services – design assessment, type approvals, third-party inspection, verification and testing, advisory and engineering services
- Certification Authentication
  - *Digital certificates* – DNV certificates that are digitally signed and for which the electronic file is available, the authenticity of the suspected certificate can be checked in the signature field or in the signature panel.
  - *Approval Finder* – Using Approval Finder is the preferred option for checking validity of certificates.
  - *Paper copy* – DNV certificates which are *only available as paper copy* (either digitally or manually signed), and verification of the above is not possible, then contact the **local DNV office** or check with the DNV issuing unit stated on the certificate.